A3: Style Transfer Report

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Introduction

Neural Style Transfer (NST) combines the content of one image with the artistic style of another using deep neural networks. This project implements the method proposed by Gatys et al. (2016) using a pre-trained VGG19 model.

Part 1: Implementation

Content is extracted from conv4_2, and style is captured using Gram matrices across multiple layers.

Part 2: Experiments

Experiments varied style weights (5e4–1e7) and learning rates (0.001–0.003).

Results (Insert Images)

Insert:

- Content Image

Higher layers of the network capture semantic content, while style is represented by correlations between feature maps using Gram matrices, as described in Gatys et al. (2016).

- Style Image

- Output Images

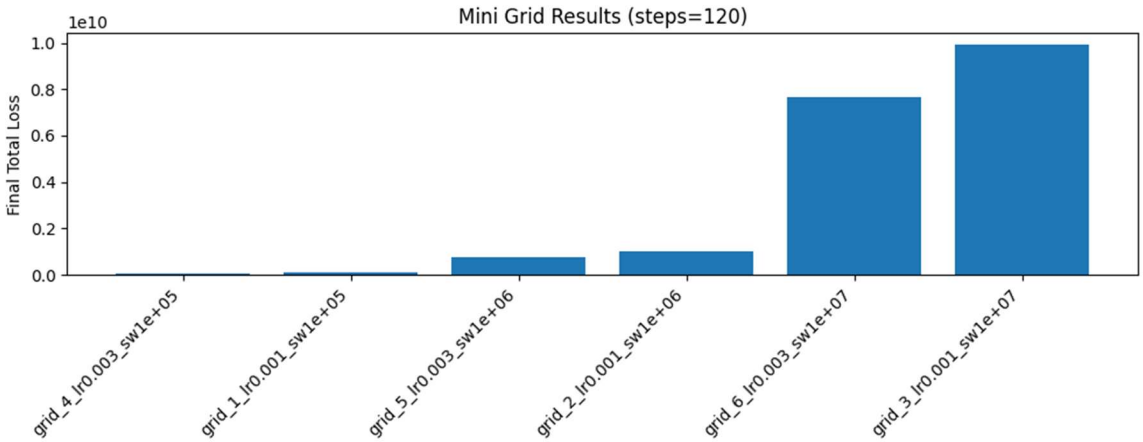
long_lr0.003_sw5e4_s600

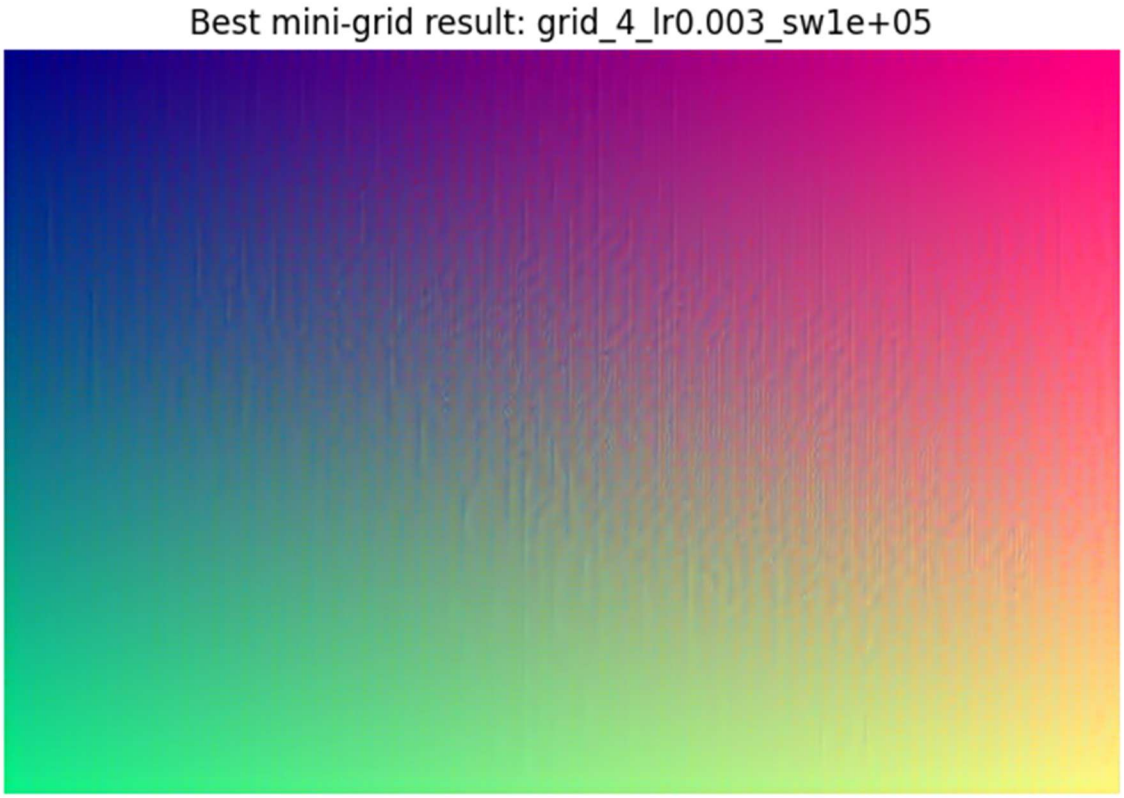
long_lr0.003_sw1e5_s600

Figure 1: With style weight 5e4, the output preserves building structure while applying mild texture patterns.

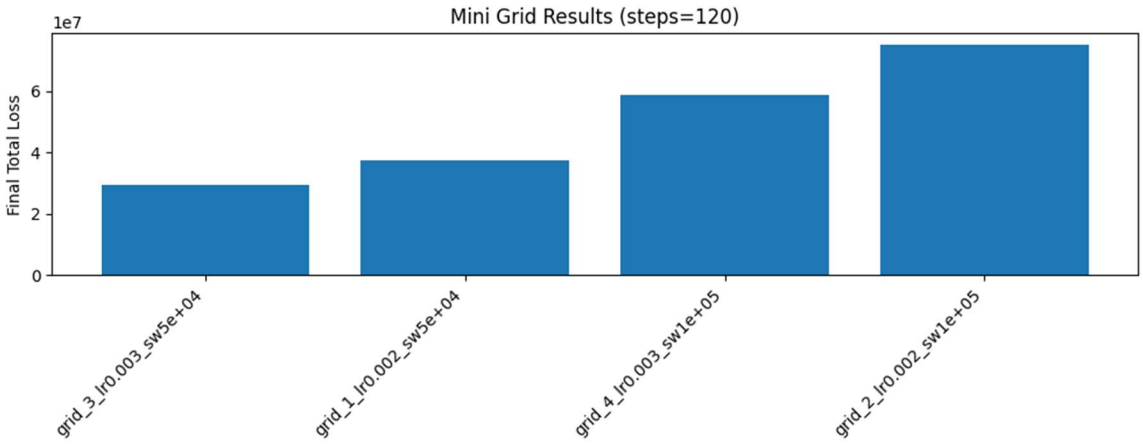
Figure 2: With style weight 1e5, stronger stylization appears, but some structural distortion occurs.

- Graphs





From the mini-grid results, higher style weights lead to higher loss values and stronger stylization. However, visual quality does not always correlate with lower loss.



Conclusion

Balanced hyperparameters produce the best visual results, preserving content while applying style.